KINETIC THEORY OF GASES · 10 PRACTICE PROBLEMS

Basic P, V, T Graphs

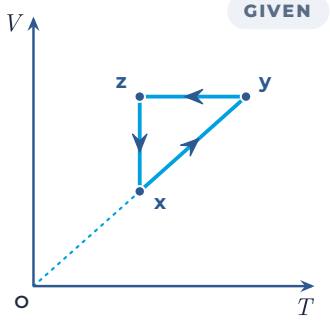
Try all 10 first. Stuck on a concept? The theory notes start on page 4. Answers and short solutions are on the last page.

Q1 JEE Main 2020 · 8 Jan, Shift 1

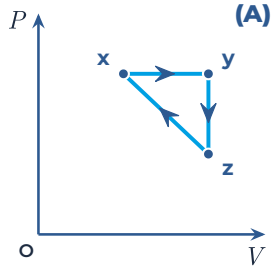
A thermodynamic cycle $xyzx$ is shown on a V–T diagram. The P–V diagram that best describes this cycle is:

(Diagrams are schematic and not to scale.)

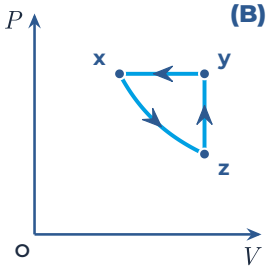
GIVEN



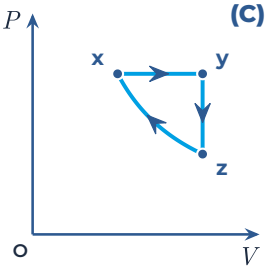
(A)



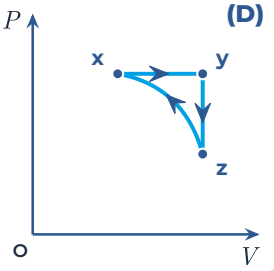
(B)



(C)



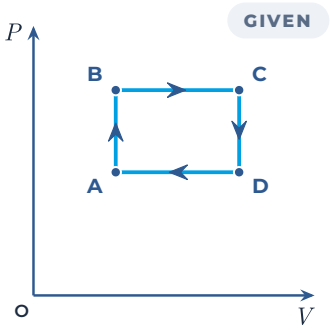
(D)



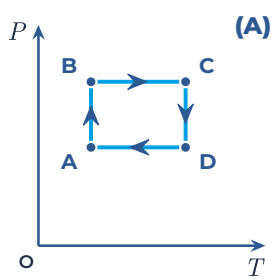
Q2

A fixed mass of an ideal gas is taken through the rectangular cycle **ABCDA** on the P–V diagram (AB and CD vertical, BC and DA horizontal). Which P–T diagram represents the same cycle?

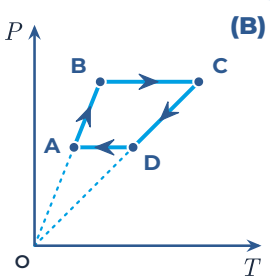
GIVEN



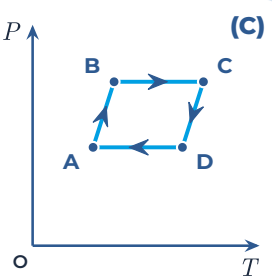
(A)



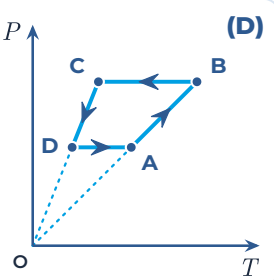
(B)



(C)



(D)



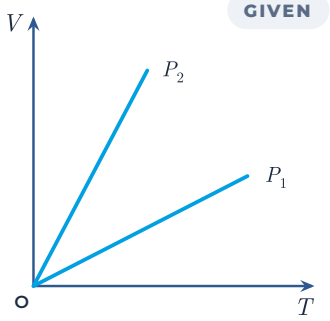
Q3 JEE Main 2024 · 31 Jan, Shift 1

The given figure represents two isobaric processes for the same mass of an ideal gas. Then

(A) $P_2 \geq P_1$ **(B)** $P_2 > P_1$

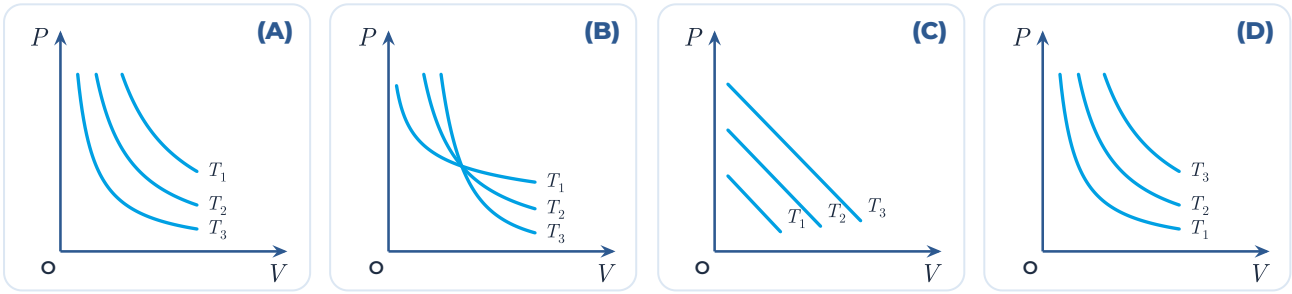
(C) $P_1 = P_2$ **(D)** $P_1 > P_2$

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Q4 JEE Main 2023 · 24 Jan, Shift 2

In an isothermal change, the change in pressure and volume of a gas can be represented for three different temperatures $T_3 > T_2 > T_1$ as:



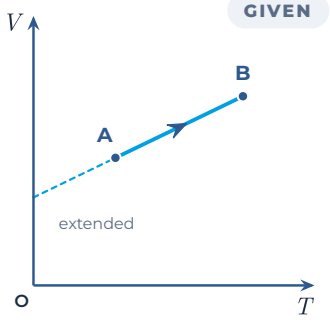
Q5

A fixed mass of an ideal gas goes from A to B along a straight line on the V–T graph. Extended backwards, the line meets the V-axis **above** the origin. During A → B, the pressure of the gas

(A) remains constant **(B)** decreases

(C) increases **(D)** first decreases, then increases

GIVEN



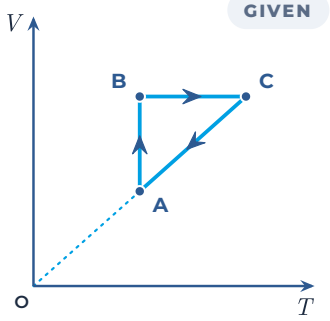
Q6 JEE Main 2025 · 29 Jan, Shift 1

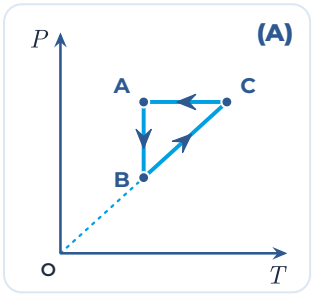
A container of fixed volume contains a gas at 27 °C. To double the pressure of the gas, the temperature of the gas should be raised to _____ °C. *(Numerical answer)*

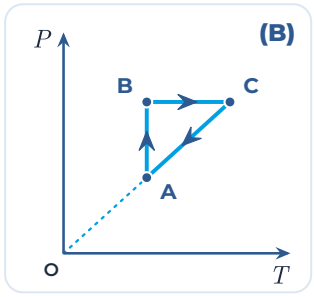
Q7

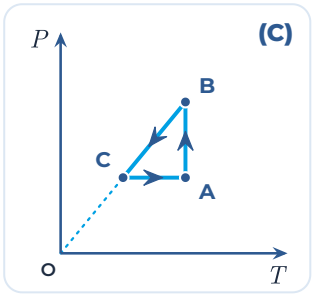
A fixed mass of an ideal gas is taken through the cycle **ABCA** shown on the V–T graph (AB vertical, BC horizontal, CA on a line through O). Which P–T graph represents the same cycle?

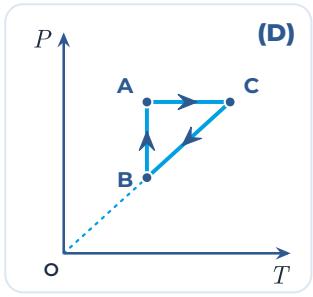
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Q8 JEE Main 2025 · 3 Apr, Shift 2

Pressure of an ideal gas, contained in a closed vessel, is increased by 0.4% when heated by 1 °C. Its initial temperature must be:

(A) 25 °C **(B)** 2500 K **(C)** 250 K **(D)** 250 °C

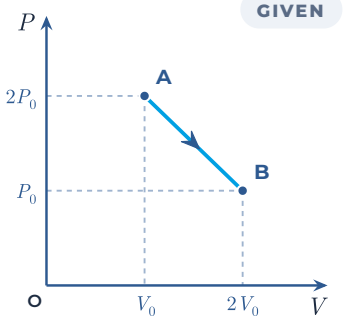
Q9 JEE Main 2016 · Offline

n moles of an ideal gas undergo the process A → B shown in the figure (a straight line on the P–V diagram). The maximum temperature of the gas during the process will be:

(A) $\frac{9P_0V_0}{2nR}$ **(B)** $\frac{9P_0V_0}{nR}$

(C) $\frac{9P_0V_0}{4nR}$ **(D)** $\frac{3P_0V_0}{2nR}$

GIVEN



Q10 JEE Main 2021 · 31 Aug, Shift 1

For an ideal gas, the instantaneous change in pressure p with volume v is given by $\frac{dp}{dv} = -ap$. If $p = p_0$ at $v = 0$ is the given boundary condition, then the maximum temperature one mole of gas can attain is (R is the gas constant):

(A) $\frac{p_0}{aeR}$ **(B)** $\frac{ap_0}{eR}$ **(C)** infinity **(D)** 0 °C

THEORY · KINETIC THEORY OF GASES

Gas Processes & Their Graphs

Hold one of P, V, T fixed and the other two get locked together. Learn the three shapes once, and any cycle can be redrawn on any pair of axes.

One equation behind every graph

$$PV = nRT$$

For a fixed amount of gas, n and R never change.

- **Isothermal** — T constant → $PV = \text{const}$
- **Isobaric** — P constant → $V \propto T$
- **Isochoric** — V constant → $P \propto T$

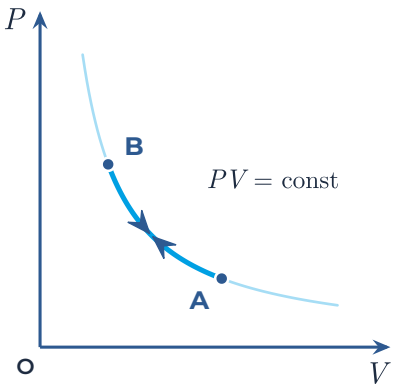
Each process keeps the same colour in every graph of these notes.

1 Isothermal Process

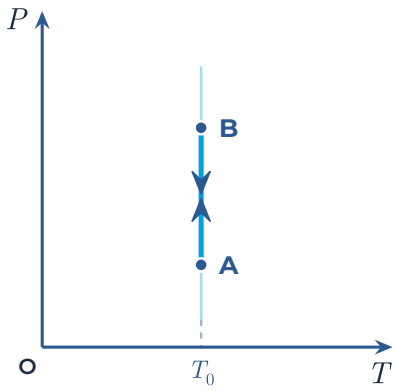
T = constant

$PV = nRT$ *n, R, T all constant* $\Longrightarrow PV = \text{constant}$ *Boyle's law*

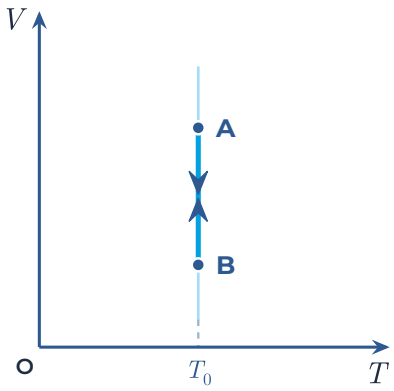
EXAMPLE Push a piston in **slowly** while the cylinder sits in a large water bath. Heat flows out into the bath, so T never changes.



P–V · rectangular hyperbola



P–T · vertical line at T_0



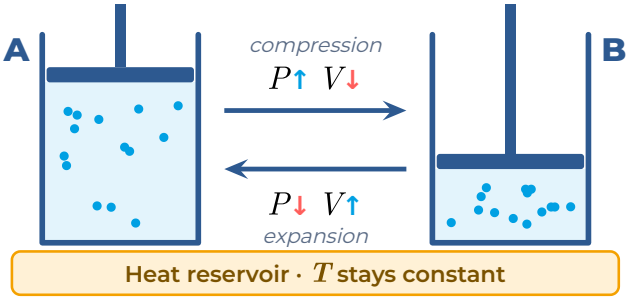
V–T · vertical line at T_0

A → B $P\uparrow$ $V\downarrow$ *compression*

B → A $P\downarrow$ $V\uparrow$ *expansion*

Read the arrows: the pair →← on a line means the gas can go either way, A → B or B → A.

Two isotherms? The hotter one ($PV = nRT$ larger) lies farther from the origin.

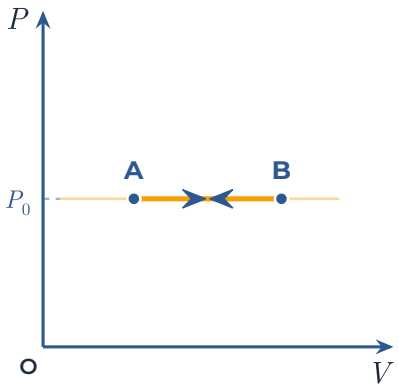


2 Isobaric Process

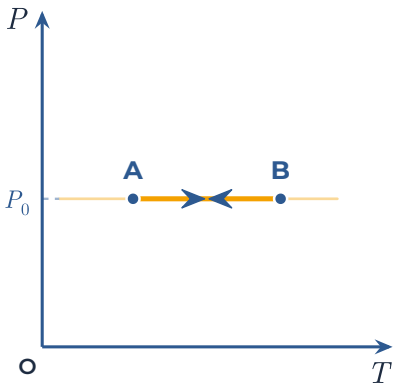
P = constant

$PV = nRT$ $\Longrightarrow V = \frac{nR}{P}\,T$ $\frac{nR}{P}$ *is constant* $\Longrightarrow V \propto T,\ V = kT$ *Charles' law*

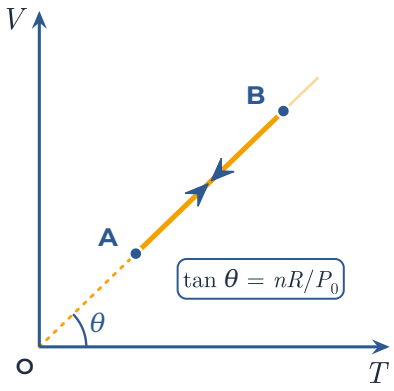
EXAMPLE Heat a gas under a **freely moving** piston. The piston rises and the gas expands, but the pressure on it stays the same.



P–V · horizontal line at P_0



P–T · horizontal line at P_0



V–T · straight line through O

A → B $V\uparrow\ T\uparrow$ *heated, gas expands* **B → A** $V\downarrow\ T\downarrow$ *cooled, gas contracts*

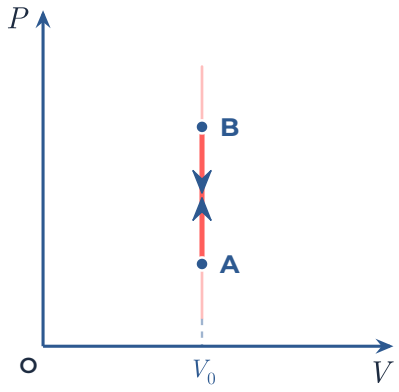
Slope: $\tan\theta = \frac{nR}{P_0}$. Lower pressure → steeper V–T line.

3 Isochoric Process

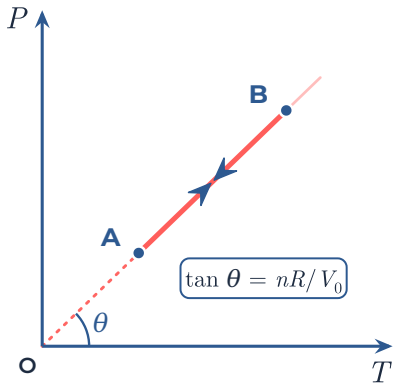
V = constant

$PV = nRT$ $\Longrightarrow P = \frac{nR}{V}\,T$ $\frac{nR}{V}$ *is constant* $\Longrightarrow P \propto T,\ P = kT$ *Gay-Lussac's law*

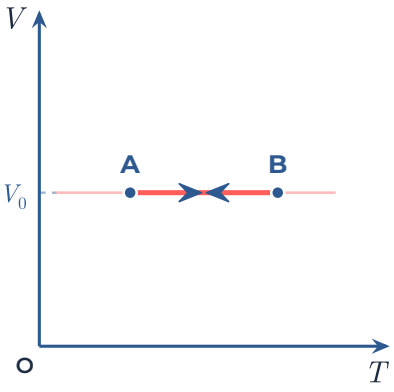
EXAMPLE Heat a gas in a **sealed rigid** container, like a closed pressure cooker. The volume cannot change, so the pressure climbs.



P–V · vertical line at V_0



P–T · straight line through O



V–T · horizontal line at V_0

A → B $P\uparrow\ T\uparrow$ *heated in a rigid vessel* **B → A** $P\downarrow\ T\downarrow$ *cooled in a rigid vessel*

Slope: $\tan\theta = \frac{nR}{V_0}$. Smaller volume → steeper P–T line.

At a Glance

Process	Constant	Relation	P–V graph	P–T graph	V–T graph
Isothermal	T	$PV = \text{const}$	Rectangular hyperbola	Vertical line	Vertical line
Isobaric	P	$V \propto T$	Horizontal line	Horizontal line	Line through O
Isochoric	V	$P \propto T$	Vertical line	Line through O	Horizontal line

RULE 1
The **constant** quantity on an axis gives a line **parallel to the other axis**.

RULE 2
Two quantities in **direct proportion** give a **straight line through O**.

RULE 3
$PV = \text{const}$ gives a **rectangular hyperbola** on the P–V graph.

Redrawing a cycle in 4 steps

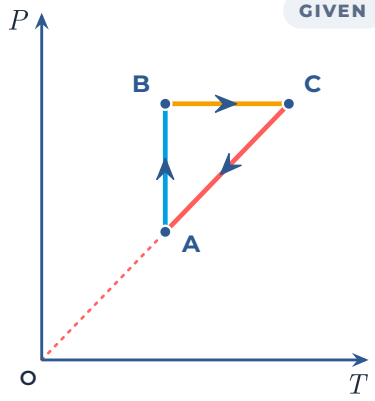
1. **Name each step.** Vertical or horizontal line → that quantity is constant. Line through O on P–T → V constant; on V–T → P constant.
2. **Write what rises and falls** in each step ($P\uparrow$ $V\downarrow$ …).
3. **Place the points** on the new axes in the right order (e.g. $V_B < V_A$).
4. **Join them** with the shape from the table and keep the arrows in the same order A → B → C …

Common slips to avoid

- An isothermal on the P–V graph is a **curve** (hyperbola), never a straight line.
- A straight line on P–T or V–T means isochoric / isobaric **only if it passes through O**. Extend it dotted to check.
- Keep the **same order of arrows** (A → B → C …) on every graph you redraw.

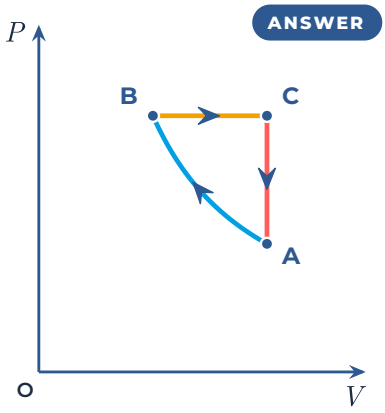
Ex 1 For the cycle **A → B → C → A** shown on the P–T graph, draw the P–V and V–T graphs.

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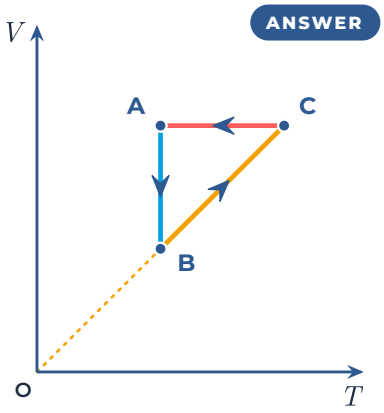
Given · P–T

ANSWER



P–V

ANSWER



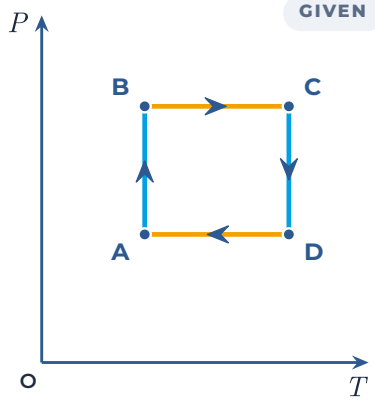
V–T

A → B	Isothermal	T const · $P\uparrow$ $V\downarrow$
B → C	Isobaric	P const · $T\uparrow$ $V\uparrow$
C → A	Isochoric	V const · $P\downarrow$ $T\downarrow$

KEY
C → A lies on a line through O on the P–T graph, so P/T is constant → V is constant. Hence $V_C = V_A$: a vertical drop on P–V, a horizontal line on V–T.

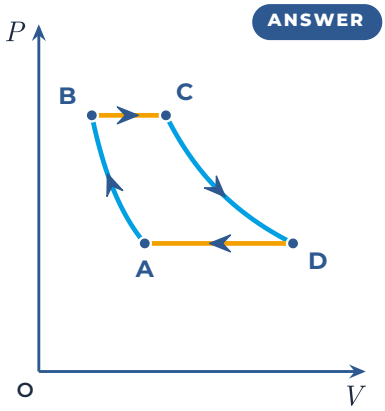
Ex 2 For the cycle **A → B → C → D → A** shown on the P–T graph, draw the P–V and V–T graphs.

GIVEN



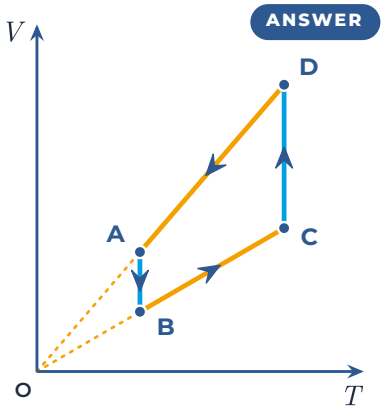
Given · P–T

ANSWER



P–V

ANSWER



V–T

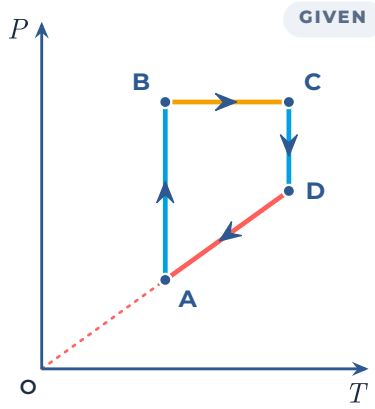
A → B	Isothermal	T const · $P\uparrow$ $V\downarrow$
B → C	Isobaric	P const · $T\uparrow$ $V\uparrow$
C → D	Isothermal	T const · $P\downarrow$ $V\uparrow$
D → A	Isobaric	P const · $T\downarrow$ $V\downarrow$

KEY

C → D is the hotter isotherm, so its hyperbola sits farther from O. On V–T, D → A is at the lower pressure, so its line through O is steeper than B → C.

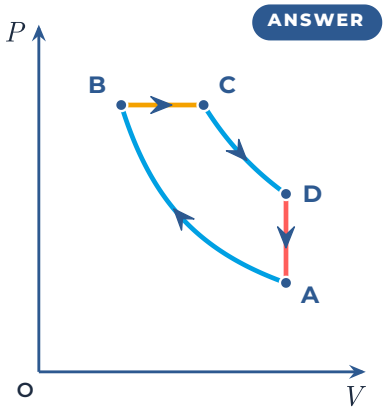
Ex 3 For the cycle **A → B → C → D → A** shown on the P–T graph, draw the P–V and V–T graphs.

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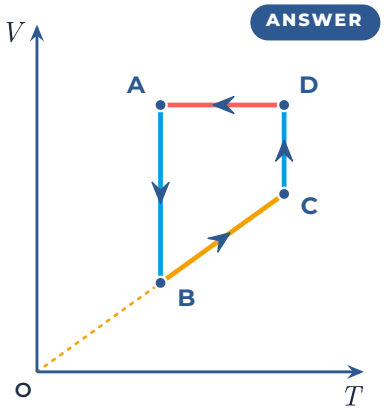
Given · P–T

ANSWER



P–V

ANSWER



V–T

A → B	Isothermal	T const · $P\uparrow$ $V\downarrow$
B → C	Isobaric	P const · $T\uparrow$ $V\uparrow$
C → D	Isothermal	T const · $P\downarrow$ $V\uparrow$
D → A	Isochoric	V const · $P\downarrow$ $T\downarrow$

KEY

D → A lies on a line through O on the P–T graph, so $V_D = V_A$. On P–V it is a vertical drop from D to A; on V–T it is a horizontal line.

✓ Answer Key & Solutions

Q1 **(C)**

$x \to y$: P const → horizontal line. $y \to z$: V const, $T\downarrow$ → P drops, vertical line. $z \to x$: T const → hyperbola, curving towards O.

Q2 **(B)**

AB, CD: V const → lines through O. AB has the smaller V, so it is steeper. BC, DA: P const → horizontal lines.

Q3 **(D)**

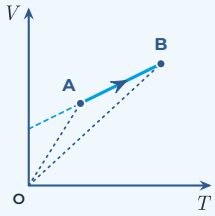
Slope of a V–T line = $\frac{nR}{P}$. Steeper line → lower pressure. So $P_1 > P_2$.

Q4 **(D)**

Same V, higher T → higher P. So the T_3 curve is farthest from O. Isotherms never cross.

Q5 **(C)**

$P \propto T/V$. Line OB is less steep than OA, so V/T falls and P rises.



Q6 **327**

Use kelvin: 300 K × 2 = 600 K = **327 °C** (not 54 °C).

Q7 **(A)**

AB: T const, $V\uparrow$ → $P\downarrow$, vertical drop. BC: V const → line through O. CA: P const → horizontal line.

Q8 **(C)**

V const → $\frac{\Delta P}{P} = \frac{\Delta T}{T}$. $0.004 = \frac{1}{T}$ → $T = 250$ K.

Q9 **(C)**

$T \propto PV$. On line AB, PV is largest at the middle point, where $V = \frac{3V_0}{2}$ and $P = \frac{3P_0}{2}$. So $T_{max} = \frac{9P_0V_0}{4nR}$.

Q10 **(A)**

Solving gives $p = p_0 e^{-av}$. $T = pv/R$ is largest at $v = 1/a$, so $T_{max} = \frac{p_0}{aeR}$.

PYQ sources cross-checked on Cracku, Competishun, ExamSIDE and IIT Kanpur SATHEE. Figures are redrawn; the options for Q1 and Q4 are redrawn too, so their letters may differ from the original paper.